Assignment 9: Advanced SQL Queries Using Joins and Subqueries

COLLEGE MANAGEMENT SYSTEM

DBMS Assignment 9

Database: CollegeManagementSystem

SQL Platform: MySQL 8.0.16 or later

Project scope: departments, faculty, students, courses, and enrollments.

Note: Query outputs described in this report are expected results from the included fictional sample data. Run the SQL script in MySQL Workbench and capture actual screenshots if your instructor requires them.

Report 1: Students and departments

```
SELECT s.student_name,s.register_no,d.department_name
FROM Students s JOIN Departments d ON s.department_id=d.department_id
ORDER BY d.department_name,s.student_name;
```

Report 2: Courses and assigned faculty

```
SELECT c.course_code,c.course_name,f.faculty_name,d.department_name
FROM Courses c JOIN Faculty f ON c.faculty_id=f.faculty_id
JOIN Departments d ON c.department_id=d.department_id ORDER BY c.course_code;
```

Report 3: Enrollment details

```
SELECT s.student_name,c.course_name,e.semester,e.academic_year,e.grade
FROM Enrollments e JOIN Students s ON e.student_id=s.student_id
JOIN Courses c ON e.course_id=c.course_id ORDER BY s.student_name,c.course_name;
```

Report 4: Number of courses per student

```
SELECT s.student_name,COUNT(e.enrollment_id) AS courses_taken
FROM Students s LEFT JOIN Enrollments e ON s.student_id=e.student_id
GROUP BY s.student_id,s.student_name ORDER BY courses_taken DESC;
```

Report 5: Students with A+ grade (IN subquery)

```
SELECT student_name FROM Students
WHERE student_id IN (SELECT student_id FROM Enrollments WHERE grade='A+');
```

Report 6: Faculty earning above average

```
SELECT faculty_name,salary FROM Faculty
WHERE salary > (SELECT AVG(salary) FROM Faculty) ORDER BY salary DESC;
```

Report 7: Student count by department

```
SELECT d.department_name,COUNT(s.student_id) AS student_count
FROM Departments d LEFT JOIN Students s ON d.department_id=s.department_id
GROUP BY d.department_id,d.department_name ORDER BY student_count DESC;
```

Explanation

INNER JOIN combines rows with matching keys. LEFT JOIN keeps all rows from the left table even when no matching row exists. IN uses values returned by a subquery. The salary query compares each faculty salary with the average salary.

End of Assignment Report